Experiment:- 6

Problem statement:-

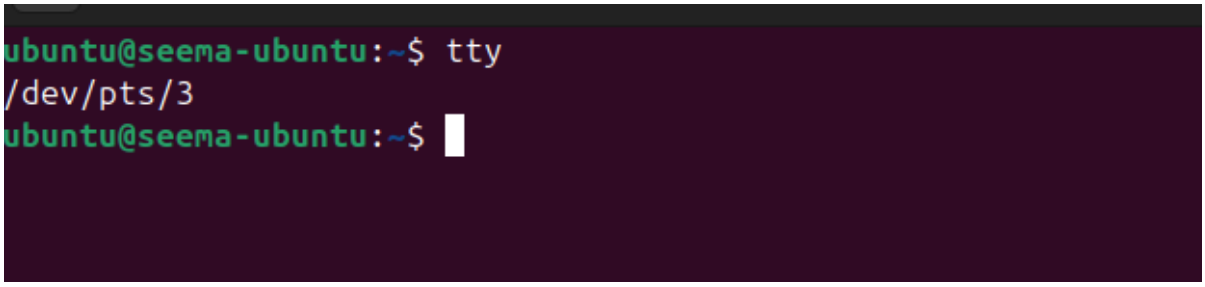
Linux operating systems support multiple user sessions and interfaces through the use of virtual terminals and graphical environments. Understanding how Linux manages text based terminals, graphical display servers, display managers, X clients, and window managers is essential for system administration, troubleshooting, and cyber security operations.

The problem is to explore and understand virtual terminals, display managers, X clients, and window managers, and to practice switching between text-based and graphical interfaces in a Linux environment.

Aim:- Understanding virtual terminal, display manager, X clients, and window manager in linux.

Procedure:-

1. Log in to the Linux system and open the terminal. Log in to the Linux system. Open the terminal. Check the current terminal: tty



2. Switch to another virtual terminal using: Ctrl + Alt + F2. Log in and verify terminal access. Switch back to graphical mode: Ctrl + Alt + F1 or Ctrl + Alt + F7

3. Check logged-in users: who w

systemctl statuwho

```
/dev/pts/3
ubuntu@seema-ubuntu:~$ who
ubuntu   seat0        2026-03-30 11:02 (login screen)
ubuntu   :0           2026-03-30 11:02 (:0)
ubuntu@seema-ubuntu:~$
```

4. Check the display manager status: systemctl status lightdm

```
ubuntu@seema-ubuntu:~$ systemctl status lightdm
Unit lightdm.service could not be found.
ubuntu@seema-ubuntu:~$ systemctl status lightdm
Unit lightdm.service could not be found.
ubuntu@seema-ubuntu:~$
```

5. Verify X session variable: echo $DISPLAY

```
Unit lightdm.service could not be found.
ubuntu@seema-ubuntu:~$ echo $DISPLAY
:0
ubuntu@seema-ubuntu:~$
```

6. List running X-related processes: ps -e | grep X

```
ubuntu@seema-ubuntu:~$ ps -e |grep x
     19 ?        00:00:00 rcu_exp_par_gp_kthread_worker/1
     20 ?        00:00:00 rcu_exp_gp_kthread_worker
   1201 ?        00:00:53 irq/16-vmwgfx
   2609 tty2     00:00:00 gdm-x-session
   2866 ?        00:00:02 mutter-x11-fram
   3014 ?        00:00:00 xdg-permission-
   3137 ?        00:00:00 gsd-xsettings
   3533 ?        00:00:08 ibus-extension-
   3541 ?        00:00:00 ibus-x11
   3770 ?        00:00:01 xdg-desktop-por
   3777 ?        00:00:00 xdg-document-po
   3789 ?        00:00:01 xdg-desktop-por
   3844 ?        00:00:00 xdg-desktop-por
   5131 ?        00:00:00 kworker/R-xfsalloc
   5132 ?        00:00:00 kworker/R-xfs_mru_cache
ubuntu@seema-ubuntu:~$
```

Expected Output

- Successful switching between virtual terminals.
- Identification of display manager and X session.
- Display of active user sessions.
- Confirmation of graphical environment variables.

Result

Virtual terminals, display managers, X clients, and window managers were successfully explored. The student gained practical understanding of Linux system interfaces and graphical session management.